process. Spoken communication starts with a source or speaker who has both a purpose for engaging in communication and the ability to properly encode and articulate the thought to be conveyed. The actual thought to be expressed is termed the "message." The message is made up of auditory stimuli organized in meaningful linguistic units. Visual and tactile cues are also provided by the speaker in conjunction with the production of the auditory message. Another critical component of the process is the auditory feedback provided to the speaker while producing speech, which then provides an opportunity for any needed adjustments or corrections to occur. The communication situation in which the message is conveyed is referred to as the "environment." Factors associated with the environment may include competing background noise, the number of speakers, amount of light in a room, and reverberation. These factors can drastically alter the amount and quality of the communication that takes place, not to mention the effort of the listener. The final major component of the communication process is the receiver or listener, who is charged with the responsibility of receiving and properly decoding and interpreting the speaker's intended message. The listener also provides additional feedback to the speaker about how the message is being received.

These basic components of the spoken communication process and their sequence are found in Figure 4–1. All the major components are equally important in accomplishing the desired end: effective communication. Disruption or elimination of any one part may result in partial or complete failure of the communication process. Proper application of this communication model is of concern to us throughout the chapter and the entire book.

Auditory Perception

Our ability to communicate verbally with others depends to a great extent on the quality of our auditory perception of the various segmental (individual speech sounds) and suprasegmental (rate, rhythm, intonation) elements that make up speech. The following sections focus on the basic aspects of auditory perception: the intensity, frequency, and duration components of speech and transitional cues. The impact of hearing loss on speech perception is also discussed.

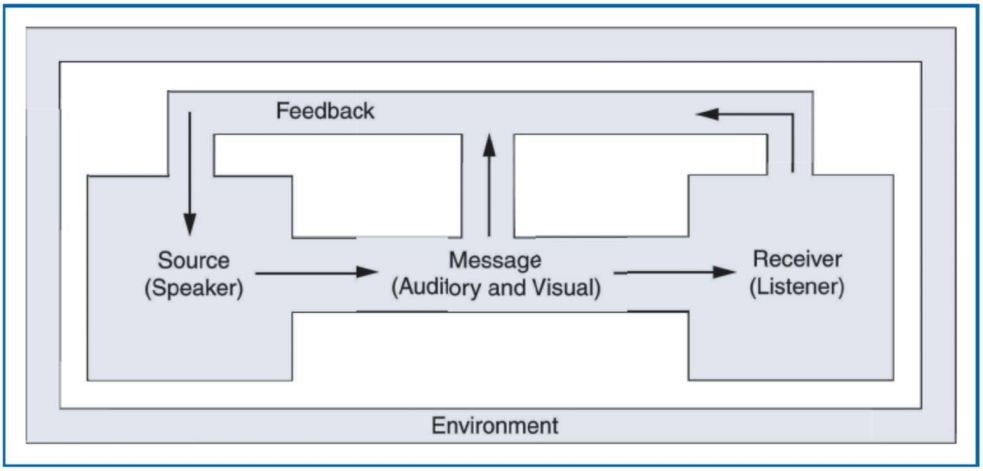


FIGURE 4–1. A simple model of the oral, or verbal, communication process.

Development of Auditory Skills

The normal development of speech and language is heavily dependent on the infant's acquisition of auditory skills.

Detection simply involves knowing that a sound is present, whereas discrimination is the ability to distinguish when two separate sounds are different.

It is both important and amazing to realize that the unborn infant possesses a functional auditory system that allows the child to begin perceiving auditory stimuli prior to birth. This is followed by further development and refinement in the neonate's auditory-processing skills in the days and weeks immediately following birth. As a result, the newborn infant not only can *detect* auditory stimuli (i.e., knowing a sound is there) but also can make gross discriminations (i.e., tell the difference) between various auditory signals on the basis of frequency and intensity parameters. This process of selective listening is extended to speech stimuli within a few weeks following birth. The rather rapid emergence of auditory skills, as described by Northern and Downs (2014), is crucial for the development of speech-processing abilities as well as for the development of speech and language in the infant. The development of auditory and speech-language skills can be seriously impacted with hearing loss, making newborn hearing screening and early intervention particularly important for spoken language development.

Basic Perception Abilities

Although the human auditory system has sophisticated perceptual capabilities, it is limited, to some extent, in the signals it can process. Optimally, the normal human ear is capable of perceiving auditory signals comprising frequencies ranging from about 20 to 20,000 Hz. Stimuli made up entirely of frequencies below and above these limits cannot be detected. Intensity limits, as shown in Figure 4–2, vary as a function of the frequency of the auditory stimulus. The maximum range of intensity we are capable of processing occurs at 3000 to 4000 Hz and varies from about 0 dB SPL to approximately 130 to 140 dB SPL. Signals with intensity

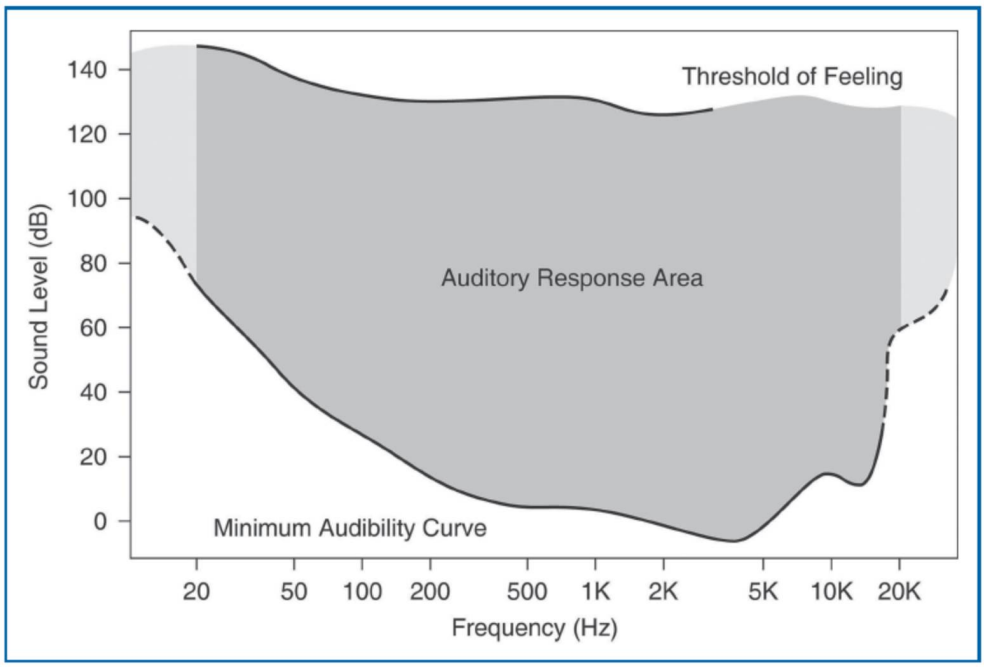


FIGURE 4–2. The auditory response area for persons with normal hearing. Adapted with permission from Durrant & Lovrinic (1995).

of less than 0 dB SPL are generally not perceived; in contrast, signals in excess of 130 to 140 dB SPL produce the sensations of feeling and pain rather than hearing.

In addition to detecting acoustic signals, the human ear can also discriminate different stimuli on the basis of only minor differences in their acoustic properties. Our ability to discriminate changes in the frequency, intensity, or duration of a signal is influenced by the magnitude of each of the other factors. Stevens and Davis (1938) estimated that the normal ear is capable of perceiving approximately 340,000 distinguishable tones within the audible range of hearing. This total number was based only on frequency and intensity variations of the stimuli, but it suggests that our auditory system possesses amazing discrimination powers.

Acoustics of Speech

Knowledge about the acoustic properties of speech is important for understanding how speech is perceived. Therefore, basic information relevant to this process is covered in the following sections.

Intensity Parameters of Speech

The normal human ear is capable of processing signals within an intensity range approaching 130 dB; however, the range of intensity normally found in speech is relatively small. The average intensity of speech, when measured at a distance of approximately 1 meter from the speaker, approximates 65 dB SPL. This corresponds to a value of about 45 dB HL when expressed audiometrically. The average shout will approach 85 dB SPL (65 dB HL), and faint speech occurs at about 45 dB SPL (25 dB HL). Thus, a potential range of about 40 dB exists between the average intensities found within the softest and loudest speech we are exposed to in common communication situations. Distance between the speaker and listener can influence the actual intensity levels for a given communication situation.

Considerable variability exists in the acoustic energy normally associated with individual speech sounds. Table 4–1 lists the relative phonetic powers of the phonemes, as reported by

TABLE 4–1. Relative Phonetic Power of Speech Sounds as Produced by an Average Speaker

ɔ	680	l	100	t	15
ɑ	600	ʃ	80	g	15
ʌ	510	ŋ	73	k	13
æ	490	m	52	v	12
ʊ	460	ʧ	42	ð	11
ɛ	350	n	36	b	7
u	310	ʤ	23	d	7
ɪ	260	ʒ	20	p	6
i	220	z	16	f	5
r	210	s	16	θ	1

Vowels are produced with greater intensity than consonants.

Fletcher (1953). As illustrated, the most powerful phoneme, /ɔ/, possesses an average of about 680 times as much energy as the weakest phoneme, /θ/, representing an average overall difference in intensity between the two speech sounds of approximately 28 dB. Since a considerable amount of variability also exists in the intensity of individual voices, Fletcher estimated that, collectively, different speakers may produce variations in the intensity of these two phonemes as great as 56 dB. The relative power of vowels, according to Fletcher, is significantly greater than that of consonants, with the weakest vowel, /i/, having more energy than the most powerful consonant.

Frequency Parameters of Speech

The overall spectrum of speech, as seen in Figure 4–3, is composed of acoustic energy from approximately 50 to 10,000 Hz (Denes & Pinson, 1993; Neel, 2010). Closer examination of this figure also reveals that the greatest amount of energy found in speech generally is associated with frequencies below 1000 Hz. Above this frequency region, the energy of speech decreases at about a 9 dB/octave rate. The concentration of energy in the lower frequencies can be attributed largely to the fundamental frequency of the adult human voice (males, 130 Hz; females, 260 Hz) and the high intensity and spectral characteristics associated with the production of vowels. As shown in Figure 4–3, the greatest energy for speech from biological males is around 500 Hz. This is due to resonance in the vocal tract rather than just fundamental frequency. It should be noted that the fundamental frequency of children is substantially higher than that of adults, around 400 Hz (Hegde, 2010). As a result, the major energy concentration for this age group occurs higher on the frequency scale than for adults.

A key aspect of speech production and perception is the information associated with the segmental elements of speech. The segmental components consist of the numerous features

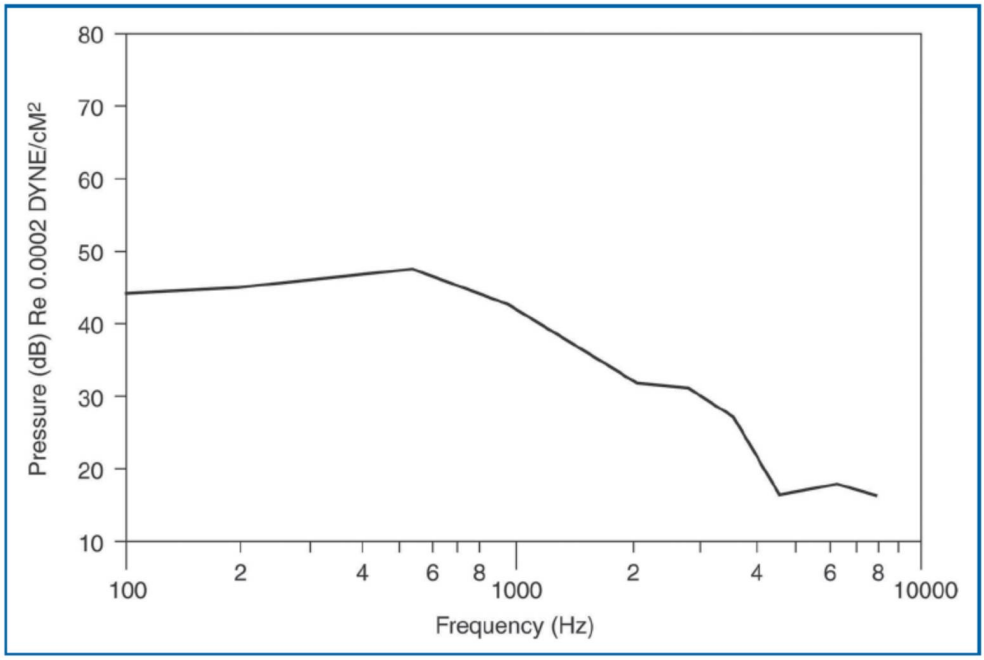


FIGURE 4–3. Long-interval acoustic spectrum of male voices. Measurement made with microphone 18 inches from speaker's lips. Source: Miller (1951).

associated with the individual vowel and consonant phonemes of the language. The vowels in English are composed mainly of low- and midfrequency energy and, as indicated earlier, contribute most of the acoustic power in speech. Specifically, the frequency spectrum of each vowel contains at least two or three areas of energy concentration that result from the resonances that occur in the vocal tract during phonation. These points of peak amplitude are referred to as formants, and their location and pattern on the frequency continuum are unique for each vowel. Figure 4–4 illustrates the approximate location of the major formants associated with the vowels, as spoken by adults who are biologically male. Formants provide important acoustic cues for the identification of vowels. However, it is important to note that even though each of the vowels has several formants, we need to hear only the first two or three to be able to accurately perceive the vowel spoken (for a review, see Kent & Vorperian, 2018; Peterson & Barney, 1952).

Formant refers to a band of frequencies that are resonated, or boosted in energy, by the vocal tract.

The consonants in English display a broader high-frequency spectral composition than do vowels. This is particularly true for those consonants that do not use voicing and whose production involves substantial constriction of the articulators. Although they contain relatively little overall energy, or intensity, when compared with vowels, consonants are extremely important in determining the intelligibility of speech. Consequently, accurate perception of consonants is vital.

Consonant perception plays a major role in comprehending speech.

Figure 4–5 contains estimates of the combined intensity and frequency values generally associated with the individual speech sounds in English. Specifically, the vertical axis presents the intensity levels in dB HL of the major components of each sound (if a particular sound has more than one major frequency component, each is noted by the same phonetic symbol), while the horizontal axis expresses the general frequency region associated with each speech

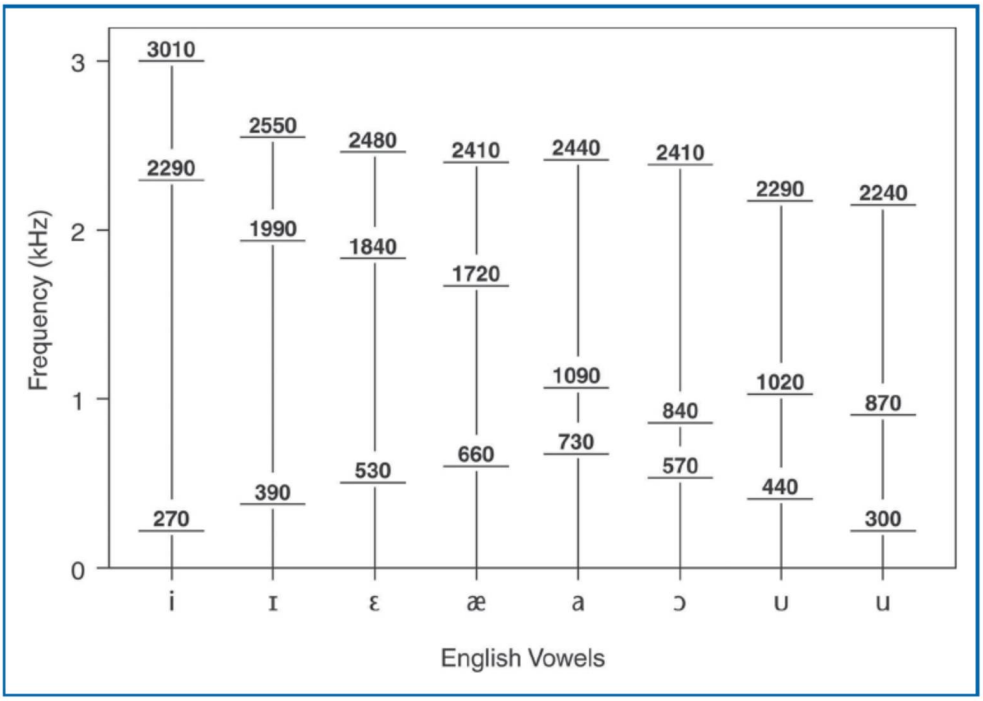


FIGURE 4–4. Mean values of formant frequencies of vowels of American English for adult males. Source: Peterson and Barney (1952).

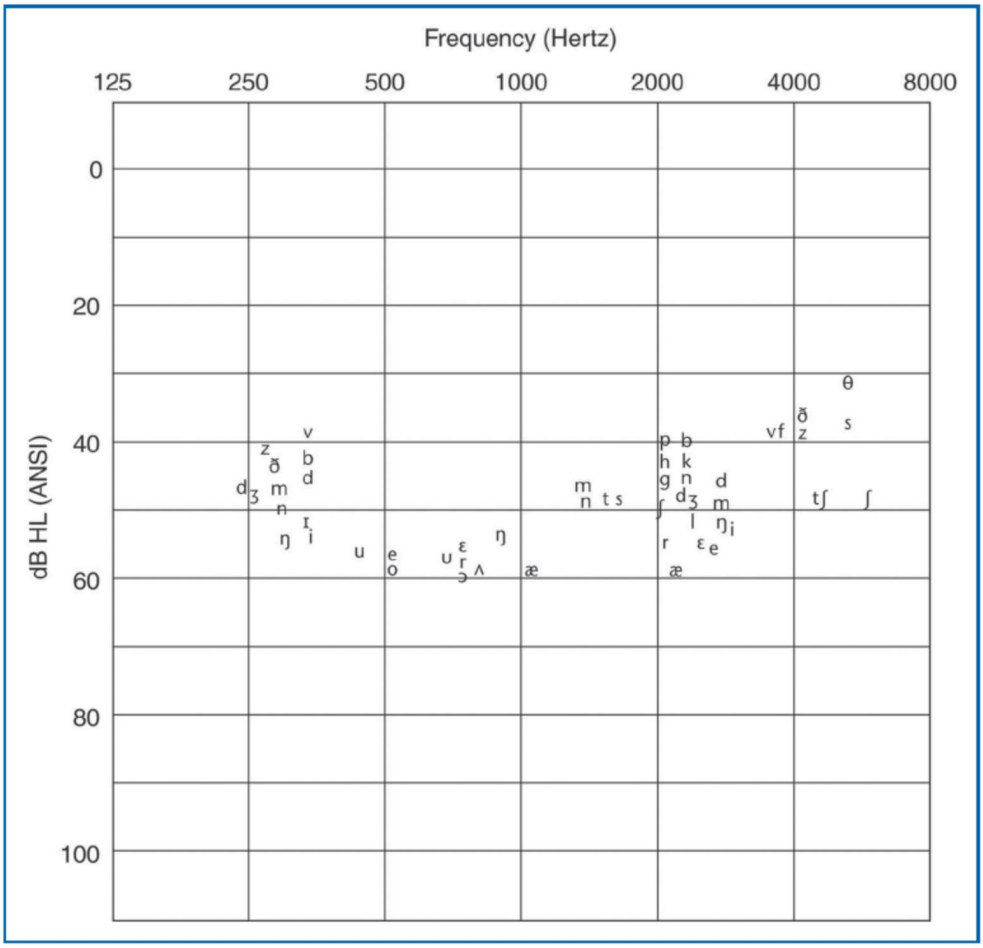


FIGURE 4–5. Intensity and frequency distribution of speech sounds in the English language. The values given should be considered only approximations and are based on data reported by Fletcher (1953) and Ling and Ling (1978). Sounds with more than one major component appear in more than one location in the figure.

sound. A close inspection of this figure discloses, as indicated earlier, that the vowels can generally be characterized as having considerable acoustic energy, for the most part confined to the low- and midfrequency range. On the other hand, the consonants demonstrate decidedly less intensity overall and a much more diffuse frequency distribution as a group. The voiced consonants generally possess a greater amount of low- and midfrequency energy, while the unvoiced consonants are made up of mid and high frequencies. All consonants appear in the upper portion of the figure, reflecting their weaker intensity values.

Voicing, nasality, duration, and place of articulation are key distinctive features found in speech.

In addition to the spectral properties associated with each of the consonants, it is important to identify the frequency characteristics related to the distinctive features associated with the production of these phonemes. Miller and Nicely's (1955) classification system includes five features: voicing, nasality, affrication, duration, and place of articulation. The voiced–voiceless distinction, as well as cues for nasality and affrication, is primarily carried by low-frequency energy. Information about place of articulation, on the other hand, is contained

in the higher frequencies. Table 4–2 categorizes each consonant phoneme by its place and manner of articulation and voicing features.

Another major component of speech is the *suprasegmentals*. The suprasegmental aspects are those features that are present in speech but are not confined to any single segment or phonemic element. Suprasegmental features, such as intonation, rhythm, stress, and pitch, are superimposed throughout speech as overall features. The suprasegmentals convey important information for speech perception, and this information is conveyed primarily in the low frequencies through acoustic cues associated with the fundamental frequency and other related aspects of speech.

The segmental components of speech are the features associated with the individual speech sounds. The suprasegmental components (rhythm and prosody, pitch, rate) are overall features of speech that are superimposed on phonemes and words.

Temporal Parameters of Speech

The duration of individual speech sounds in our language covers a range from about 20 to 400 msec (Fletcher, 1953; Lehiste, 1976). A number of factors can significantly influence the duration of a given phoneme, making the direct comparison of duration among phonemes difficult. However, vowels generally have a longer duration than consonants. Fletcher (1953) considered vowels to have average durations of between 130 and 360 msec, while the duration of consonants ranges from 20 to 150 msec. In spite of variations in absolute durational properties, individual phoneme duration does contribute toward speech perception. For example, Minifie (1973) pointed out that the duration of stop consonants (e.g., /p/ and /b/) varies systematically in a vowel–consonant–vowel context, with correct perception of the speech sound depending, to a degree, on the durational property of the phoneme produced.

The duration of vowels generally is longer and higher intensity than for consonants (Munson, 2018).

As we all know, the overall rate of speech differs considerably from speaker to speaker. For example, children's speech-sound durations are longer than adults' (Munson, 2001; Smith, 1978). The research of Goldman-Eisler, as discussed by Lehiste (1970), demonstrates that the average rate of speech used during connected discourse ranges from about 4.4 to 5.9 syllables per second. The normal rate of speech, as expressed in phoneme output, averages about 12 phonemes per second but can approach 20 phonemes when the speaker is excited (MacKay,

TABLE 4–2. Categorizing Consonants on the Basis of Manner and Place of Articulation and Voicing

Manner of Articulation	Place of Articulation						
	Bilabial	Labiodental	Linguadental	Alveolar	Palatal	Velar	Glottal
Plosives or Stops	p b			t d		k g	
Fricatives		f v	θ ð	s z	ʃ ʒ		h
Affricates					ʧ ʤ		
Nasal	m			n		ŋ	
Liquid				l, r			
Glide	w				j		

1987). Thus, the articulatory process is swift and capable of producing a flood of speech sounds and words that must be processed as effectively by the receiver, or listener, as they were produced by the speaker. Both of these are challenging tasks!

Transitional Cues

The intensity, frequency, and durational properties of a given phoneme spoken in isolation are altered significantly when the phoneme is produced with other phonemes in conversational speech. In conversational speech, the dynamic movements of the articulators in the production of adjacent phonemes produce acoustical byproducts, termed "transitional cues." These cues make up a large portion of the total speech signal and are very important in the perception of speech since they contain valuable information related to individual phoneme perception, especially for diphthongs and consonants.

For example, the second and third formants of vowels often contain transitions in frequency produced by the flowing movement of the articulators that signal the presence of particular consonants that immediately follow. These formant transitions occur as the vocal resonances shift during articulation of vowels and consonants, which are combined in speech. Likewise, the durational aspects of vowels in connected speech can be altered to convey information regarding the phoneme to follow. For example, a voiced consonant in the final position is often accompanied by increased duration of the vowel immediately before it. The prolonged vowel duration contributes to our perception of voicing in the consonant that follows. This is an example of why formant transitional cues are a vital part of the speech signal and of speech perception. This coarticulation can affect the sound patterns and presumed acoustic properties of sounds based on the combination occurring within words and sentences (Fowler & Saltzman, 1993; Ohala, 1993).

Transitional cues result from the influences of coarticulation of individual speech sounds when combined into words, phrases, and so on.

Speech Perception and Comprehension

Our discussion has emphasized the segmental and suprasegmental aspects that constitute speech. The organization and production of these crucial elements into a meaningful spoken message by the speaker and the accurate reception of this dynamic signal by the listener represent a highly complex, sophisticated process. However, just because the listener detects the segmental and suprasegmental elements of speech, does not mean the listener translates these sounds into proper understanding or comprehension of the message. The detection of speech by the auditory mechanism is only a first step in its perception.

In its most basic form, the perception of speech may be thought of as involving several important components, including:

- Detection: This basic aspect of auditory perception simply involves being aware of sound. The primary question asked related to detection is, "Do you hear the sound?" The answer to this question is "yes" or "no." Our ability to detect speech is influenced by our hearing acuity and the intensity level of the speech signal.
- Discrimination: Speech discrimination refers to the ability to distinguish among the individual speech stimuli (phonemes, syllables, etc.) of our language. The question asked related to discrimination is, "Do the two sounds sound the same or different?"
- Identification: The ability to repeat or label what one has heard by pointing to or naming. The identification-related question is, "What is that sound?" The larger the "set," the more challenging this question is to answer. The AR clinician can use set size as a way to support and/or challenge the client.

- Attention: A fundamental ingredient in the perception of speech relates to attending to or focusing on the speaker and the message being conveyed. The degree and quality of the listener's attention will influence how well speech is perceived.
- Memory: A key component in speech perception is the ability to retain or store verbal information for relatively brief periods or, in some instances, extended lengths of time. Memory is also fundamental to other components of speech perception and enables us to combine individual speech units for the purpose of deriving meaning from an entire verbal message rather than from each individual unit of the message.
- Closure: The perceived speech elements must be brought together into a meaningful whole. This process, termed "closure," helps a person to recognize speech even when some cues are absent, as with hearing loss.
- Comprehension: Full perception and understanding of the meaning of an auditory message. In comprehension, the question asked is, "What does that mean?"

Our task in AR is to consider how the client is currently functioning with their hearing technology and how we can systematically help them to be more successful with listening in their communicative environments.

Speech Perception and Hearing Loss

Our success in processing speech is closely related to many important factors, some of which are discussed in the next section.

Physical Properties

Information concerning the physical properties of speech is most relevant when considering the relationship between the perception of speech and hearing loss because the degree of our success in processing speech appears closely related to our ability to receive the coded acoustical information that makes up the signal.

The normal ear is well equipped to receive and process speech in most situations. Since most speech is normally presented at average intensity levels of around 45 dB HL, it is well within the sensitivity range of the normal human ear. Also, although we are capable of hearing auditory signals ranging in frequency from about 20 to 20,000 Hz, only a portion of the entire range is required for the reception of speech since speech contains energy from roughly 50 to 10,000 Hz. Consequently, in most listening conditions, those with normal hearing will experience little, if any, difficulty in adequately hearing the speech sounds found in oral communication.

The same does not hold true for persons with hearing loss. No longer are the intensity and frequency ranges of the ear with hearing loss always sufficient to provide total perception of the speech signal. Access to intensity and/or frequency may be limited such that it becomes difficult to hear specific speech sounds adequately. For example, a person with 50-dB thresholds from 2000 to 8000 Hz would have considerable difficulty perceiving the phonemes with spectral compositions that involve primarily those higher frequencies. The information in Figure 4–5 regarding the relative frequency and intensity characteristics of individual speech sounds as spoken at a typical conversation level helps in understanding why this occurs.

While factors such as type of hearing loss and test materials used can influence the outcome of investigations concerning hearing loss and the perception of phonemes, some

Consonant perception tends to be disrupted more by hearing loss than vowel perception does due to the lower frequency and increased intensity of most vowels.

general patterns of speech perception difficulties were previously observed for persons with hearing loss. Previous research explored the minimal difficulty in vowel perception (Owens et al., 1971). Only when the degree of the hearing loss is severe to profound does the perception of vowels become significantly altered (Erber, 1979). Consonant perception, however, presents a far more difficult listening task for those with hearing loss. Specifically, the phonemes /s/, /p/, /k/, /d/, and /t/ are among the most frequently missed by adults with sensorineural hearing loss (Owens, 1978) without audiologic intervention. The most common errors in consonant phoneme perception occur with the place of articulation feature, followed by manner of articulation. Errors in the perception of nasality and voice among consonants are generally far less frequent.

Previous investigations (Owens et al., 1972) confirmed that patients with high-frequency hearing loss, a sloping configuration of the audiogram, experience more difficulty perceiving /s/, /ʃ/, /tʃ/, and /dʒ/, as well as /t/ and /θ/ in the initial position only. These phonemes became increasingly difficult to hear accurately as the steepness of the sloping high-frequency hearing loss increased. Correct recognition of /s/ and the initial /t/ and /θ/ were found to be closely related to hearing sensitivity above 2000 Hz, while perception of /ʃ/, /tʃ/, and /dʒ/ was very dependent on sensitivity between 1000 and 2000 Hz. Similar findings suggested that, even with well-fit hearing technology, high-frequency phoneme perception may be difficult to achieve for individuals with hearing loss (Stelmachowicz et al., 2001).

As can be seen, the degree of difficulty in speech perception is closely related to the intensity and frequency features of speech and the intensity and frequency features of the hearing loss found on the conventional audiogram. However, difficulty with speech perception can also be influenced by other related variables that are discussed elsewhere in this chapter and throughout the text. Therefore, while the audiogram usually is our most useful single predictor of a person's speech perception abilities, other factors to be discussed in the next section must be considered as well. For cochlear implant outcomes, common prediction models can include duration of hearing loss, preoperative levels of speech perception, etiology, age of implantation, and preoperative hearing levels (Velde et al., 2021).

Redundancy and Noise

The perception of speech is a highly complex process that involves more than the acoustics of speech or the hearing abilities of the listener, even though these are important variables to be sure. Ultimately, for oral communication to be successful, sufficient information must be present in the message of interest for it to be perceived. The amount of information available for a given communication situation is closely associated with the concepts of redundancy and noise.

The redundancy of speech relates to its predictability—the greater the redundancy, the better the odds will be that a listener can guess what was said, even when they did not hear the entire message.

Conversational speech generally can be described as being highly redundant. That is, it contains information from a variety of sources that is available for a listener to use in comprehending a message, even though portions of the communication may not have been heard. The degree of redundancy in oral communication varies from one expression to the next, so the extent to which a listener can predict what was said will also vary. Basically, the more redundant a message, the more readily the listener can perceive it, especially in difficult listening situations. A number of factors present in a given communication situation can influence the amount of redundancy present, and Table 4–3 provides a list of some of these.

Among the many factors associated with the redundancy, or predictability, found in conversational speech for the listener to use for perception are syntactic, semantic, and situational constraints (Table 4–4). Syntactic, or structural, constraints relate to the predictable manner in which linguistic units are chained together according to the rules associated with

TABLE 4–3. A Partial List of Factors That Can Influence the Amount of Redundancy in Speech

Within the speaker	Compliance with the rules of the language Use of appropriate articulation, intonation, stress Size and appropriateness of the vocabulary used to convey the message
Within the message	Number of syllables, words, etc. Amount of context Frequency composition of the speech signal Intensity of the speech signal
Within the communication environment	Amount of acoustic noise Degree of reverberation Number of situational cues present that are related to the message
Within the listener	Familiarity with the rules of the language Familiarity with the vocabulary of the message Knowledge of the topic of conversation Hearing abilities

Source: Adapted from Sanders (1971).

TABLE 4–4. Examples of Linguistic Constraints Available to Enhance Speech Perception

Syntactic constraints: Refer to the fact that every language is governed by a set of grammatical rules that specifies the relationship between words used to communicate. For example, adjectives may be used to qualify nouns (as in "the ***blue*** shoes"); adjectives are not used to qualify verbs (as in "he ***blue*** ran").
Semantic constraints: Refer to the fact that the words used in a sentence are usually related to each other in a meaningful way. For example, although the sentence "Put the salt on the cloud" is syntactically correct, semantically it is highly improbable.
Situational constraints: Refer to the fact that language usually takes place within a physical and social context. Generally, the use of language bears some relationship with the context in which it is used. For example, in a stadium during a football game, it is more likely that the topic of discussion will center around sports-related activities than around religious beliefs and values.

Source: Adapted from Gagne and Jennings (2007).

acceptable English. The selection and use of phonemes and words in an utterance are strongly influenced by these rules, making it easier for the listener to predict what is to follow after having heard only the initial portion of the sentence. Such syntactic clues can be used in conjunction with another factor related to redundancy; namely, semantic constraints, which allow the listener to predict the type of vocabulary and expressions to be used based on

the general semantic content of the expression. When the topic of conversation is food, for example, the listener can expect to hear a rather restricted range of vocabulary peculiar to that particular topic. Use of this small range of words will increase the redundancy of the message, making it easier to predict what is said. Situational constraints also create redundancy. Our conversational partner, the location of the conversation, the time of day it takes place, and other similar factors all influence what we say and how we say it, which also can make conversational speech more predictable. All these types of constraints, along with other factors listed in Table 4–3, collectively produce the redundancy that makes the perception of speech easier for us all.

Noise refers to a host of factors that can reduce the amount of information present for the listener to use. In this context, noise refers to a variety of variables that can be counterproductive to communication, not only competing auditory noise. Table 4–5 provides a partial list of the potential sources of noise associated with oral communication with which the listener must contend. Each of these factors may reduce the amount of information in a spoken message, thus reducing the amount of redundancy, or predictability, that is available for the listener to use in perceiving speech.

Noise impacts both the quantity (how much someone hears) and the quality (the fidelity of the acoustic signal) of what someone hears. This can impact children even more than adults. For example, children may not have the cognitive or linguistic knowledge to fill in the gaps of a sentence presented in a noisy environment. They also may miss the meanings of words or the production of phonemes that are less acoustically salient. Even with the most advanced hearing technology, clinicians, family members, and audiologists can work together to promote listening environments conducive to spoken-language learning.

Thus, the degree of information available for the listener to use in perceiving a message is positively or negatively influenced by several related variables that are part of oral communication. For the listener, particularly one with hearing loss, the importance of each of these variables to the process of speech perception cannot be overstated. In addition, as previously stated, children with prelinguistic hearing loss, or hearing loss that exists prior to language development, need more help decoding the language than an older child or adult who lost their hearing after language had already been established.

TABLE 4–5. Some Potential Sources of Noise in Oral Communication

Within the speaker	Poor syntax Abnormal articulation Improper stress or inflection
Within the communication environment	Abnormal lighting Competing or distracting visual stimuli Competing or distracting auditory stimuli Reverberation
Within the listener	Lack of familiarity with the rules of the language Inability to identify the topic of the message Poor listening skills

Source: Adapted from Sanders (1971).